

B2B Sales Performance & Regional Analysis Dashboard

Project Summary Report

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Dataset:	Product Sales Dataset (2023-2024), Kaggle
Dataset Size:	200,000 transaction records
Tool Used:	Microsoft Excel (Formulas, PivotTables, Dashboard)
Purpose:	Sales Analytics Portfolio Project — Interview Preparation

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1. Executive Summary

This report documents an end-to-end sales analytics project built entirely in Microsoft Excel, using a real-world, 200,000-row transactional dataset ("Product Sales Dataset 2023-2024", sourced from Kaggle). The objective was to simulate the kind of regional and product performance analysis a B2B sales organization such as Kinlong would require from its sales analytics function — identifying which regions and product categories are performing well, where revenue is being lost, and what data-driven actions should follow.

The project covers the complete analytical workflow: raw data cleaning, formula-based aggregation, lookup-based target comparison, PivotTable summarization, KPI calculation, conditional-formatting-based visual flagging, and an interactive slicer-driven dashboard. In the course of building it, a genuine data quality issue was identified and resolved — the transaction date column contained a mixture of text-formatted and true date-formatted values, which is documented in detail in Section 5.

The analysis found that the **East** region is the strongest performer by both revenue and order volume, while the **South** region underperforms on both metrics simultaneously, pointing to a demand/reach problem rather than a pricing problem. **Electronics** is the leading product category by revenue, contributing more than a third of total company revenue, while **Accessories** is comparatively under-monetized. These findings and their business implications are discussed in full in Sections 6 through 12.

2. Project Objective

The purpose of this project was twofold: first, to build a genuinely useful sales performance dashboard using the core analytical toolkit expected of a Technical Sales / Sales Analytics role — conditional aggregation, lookup functions, PivotTables, and dashboarding; and second, to create a concrete, defensible portfolio artifact that demonstrates these skills using a large, realistic dataset rather than a toy example.

Specifically, the project set out to answer the following business questions:

- Which regions generate the most and least revenue, and how does that compare against target expectations?
- Which product categories contribute the most to overall revenue, and which underperform?
- Which individual products are the top revenue drivers?
- How is the customer base distributed geographically by state?
- What do the core sales KPIs (average order value, total revenue, regional/category performance ranking) look like?

3. Dataset Overview

The dataset used is the publicly available "**Product Sales Dataset (2023–2024)**" from Kaggle, containing 200,000 individual transaction records across the United States. Each row represents a single order line, with the following fields:

Field	Description
Order_ID	Unique identifier for each transaction (1 to 200,000)
Order_Date	Date the order was placed
Customer_Name	Name of the purchasing customer
City / State / Region / Country	Geographic location of the customer
Category / Sub_Category / Product_Name	Product classification hierarchy
Quantity	Units purchased in the transaction
Unit_Price	Price per unit (USD)
Revenue	Total transaction value (USD)
Profit	Profit generated from the transaction (USD)

Table 1: Structure of the raw dataset as received, prior to cleaning.

The dataset spans four U.S. sales regions (Centre, East, South, West), 47 states, and four product categories (Accessories, Clothing & Apparel, Electronics, Home & Furniture). Individual order values range from approximately \$17 to \$9,014, with a median order value of roughly \$465 and an average of approximately \$712.

4. Methodology

The project followed a structured, ten-step analytical workflow, moving from raw data to a fully interactive dashboard. Each step used a distinct Excel capability, chosen deliberately to demonstrate a broad and realistic sales-analytics skill set.

Step	Task Performed	Excel Capability Demonstrated
1	Data cleaning: header correction, blank/duplicate check, date standardization	Data cleaning & validation
2	Converted the cleaned range into a structured Excel Table	Structured data management
3	Added a calculated "Order Value Category" column using conditional logic	IF() logic
4	Aggregated revenue and order counts by Region and by Category	SUMIFS / COUNTIFS
5	Built a target-revenue reference table and pulled values via lookup	VLOOKUP / XLOOKUP
6	Built three PivotTables: revenue by region/month, top products by state	PivotTables
7	Applied red/green visual flags comparing actual vs. target revenue	Conditional Formatting
8	Calculated Average Order Value, Total Revenue, and regional growth KPIs	Aggregating KPI formulas
9	Built an interactive Dashboard sheet with charts and slicers	Charts, Slicers, Report Connections
10	Synthesized findings into business insights (this report)	Business interpretation

Table 2: Ten-step project methodology.

5. Data Quality Findings

A close inspection of the **Order_Date** column revealed a genuine, non-trivial data quality issue: the column did not use a single consistent data type. Of the 200,000 records, **121,351 rows (60.7%)** stored the date as plain text in "MM-DD-YY" format (e.g., "08-23-23"), while the remaining **78,649 rows (39.3%)** already stored the date as a native Excel date value.

An initial cleaning formula assumed a single text format and applied text-parsing functions (LEFT/MID/RIGHT combined with DATE) uniformly across the column. This produced silently incorrect results for the rows that were already true dates — manifesting as implausible years such as 2081, 2011, and 2100 in the cleaned output. The issue was caught by spot-checking the cleaned date range for out-of-bounds years.

Resolution: The cleaning formula was rewritten to first test whether a cell already contained a numeric (true date) value using ISNUMBER(), and only apply the text-parsing logic to the remaining text-formatted cells:

```
=IF(ISNUMBER(B2), B2, DATE(2000+VALUE(RIGHT(B2,2)), VALUE(LEFT(B2,2)), VALUE(MID(B2,4,2))))
```

This finding is significant beyond the technical fix: it demonstrates the practical importance of validating cleaning logic against the full range of a dataset rather than a small sample, since the first several rows inspected manually happened to be entirely text-formatted, masking the mixed-type issue until a full-column

check was performed.

6. Regional Performance Analysis

Revenue and order volume were aggregated by region using SUMIFS and COUNTIFS, and compared against an assumed target of \$40,000,000 per region (set close to the overall regional average, to create a realistic and defensible performance benchmark).

Region	Total Revenue (USD)	Order Count	Target Revenue (USD)	vs. Target
East	44,980,048.22	57,034	40,000,000	Above Target
West	36,242,841.73	55,428	40,000,000	Below Target
Centre	36,081,894.34	49,603	40,000,000	Below Target
South	25,102,960.64	37,935	40,000,000	Below Target

Table 3: Regional revenue and order volume vs. target. Sorted by revenue, highest to lowest.

East is the clear leading region, generating \$44.98 million in revenue from 57,034 orders — the highest figures on both dimensions. **South** is the clear laggard, generating only \$25.10 million from 37,935 orders — the lowest on both dimensions simultaneously.

This dual underperformance is analytically important: because South is low on *both* revenue and order count (rather than having a normal order count but low revenue per order), the underlying issue is more likely one of **demand, market reach, or sales coverage** in that region, rather than a pricing or discounting problem. If South had a normal order count but disproportionately low revenue, the more likely explanation would be excessive discounting or a lower-value product mix being sold there — that is not what the data shows.

7. Product Category Analysis

Category	Total Revenue (USD)	Order Count	% of Total Revenue
Electronics	57,485,698.06	51,230	40.4%
Home & Furniture	47,674,426.96	50,564	33.5%
Clothing & Apparel	27,134,365.30	62,298	19.1%
Accessories	10,113,254.61	35,908	7.1%

Table 4: Revenue and order count by product category (total revenue = \$142,407,744.93).

Electronics is the leading category, contributing 40.4% of total company revenue despite generating fewer orders (51,230) than **Clothing & Apparel** (62,298 orders). This indicates Electronics carries a substantially higher average order value than Clothing & Apparel, making it the more revenue-efficient category per transaction.

Accessories is the weakest category by a wide margin, contributing only 7.1% of total revenue despite having a meaningful order count (35,908). This category likely consists of lower unit-price items and may represent an opportunity for bundling, upselling, or cross-selling into higher-value categories rather than

being treated as a standalone growth focus.

8. Top Product Performance

A PivotTable was built to isolate the ten highest-revenue individual products across the entire dataset, identified using the Value Filter "Top 10" feature on the summed Revenue field.

Rank	Product Name	Total Revenue (USD)
1	Tempur-Pedic Mattress	9,061,755.86
2	Instant Pot	8,903,475.26
3	MacBook Air	7,362,516.81
4	Apple Watch	6,834,472.35
5	Apple iPhone 14	5,740,819.18
6	iPad Pro	5,574,458.89
7	KitchenAid Mixer	4,989,740.69
8	Storage Rack	4,463,941.27
9	Brooklinen Sheets	3,981,027.44
10	Samsung Galaxy S23	3,851,240.02

Table 5: Top 10 products by total revenue (combined total: \$60,763,447.77, or ~42.7% of all revenue).

Notably, the single highest-revenue product — the Tempur-Pedic Mattress — belongs to the Home & Furniture category rather than Electronics, despite Electronics being the stronger category overall. This suggests Electronics' category-level lead is driven by a broader spread of consistently strong products (Apple Watch, iPhone 14, iPad Pro, MacBook Air, Samsung Galaxy S23 all appear in the top 10), whereas Home & Furniture's strength is more concentrated in a small number of high-ticket items.

9. Geographic (Customer) Distribution Analysis

A third PivotTable was built to count distinct customer transactions by U.S. state, covering all 47 states present in the dataset.

State	Customer Order Count
California	10,272
Arizona	10,262
Illinois	5,832
New Jersey	5,780
Ohio	5,776

Table 6: Top 5 states by customer order count (total across all 47 states: 200,000).

California and Arizona stand out distinctly from the rest of the distribution, each contributing over 10,000 orders — roughly double the count of the next-highest states. This is a meaningfully different pattern from the regional breakdown in Section 6, where these two states' performance is distributed across the West and Centre regions respectively. This suggests that within otherwise moderate-performing regions, a small number of individual states are disproportionately responsible for order volume, which would be a useful starting point for more granular, state-level sales planning.

At the opposite end, **South Carolina (1,108 orders)**, **Idaho (2,071)**, and **Georgia (2,120)** recorded the lowest customer order counts among all states, indicating limited market penetration in these areas.

10. Key Performance Indicators (KPIs)

KPI	Value	Interpretation
Average Order Value (AOV)	\$712.04	Typical revenue generated per individual transaction
Total Revenue	\$142,407,744.93	Combined revenue across all 200,000 transactions
Best-Performing Region	East (\$44.98M)	Highest revenue and highest order count
Weakest-Performing Region	South (\$25.10M)	Lowest revenue and lowest order count
Best-Performing Category	Electronics (40.4% of revenue)	Highest revenue share, efficient per-order value
Weakest-Performing Category	Accessories (7.1% of revenue)	Lowest revenue share among all categories

Table 7: Summary of core sales KPIs derived from the dataset.

11. Dashboard Design

All findings above are consolidated into a single interactive **Dashboard** sheet within the Excel workbook, designed for at-a-glance review by a sales manager. The dashboard includes:

- A **clustered bar chart** comparing total revenue across all four regions.
- A **line chart** showing the monthly revenue trend, broken out by region, sourced from the Revenue-by-Region-by-Month PivotTable.
- A **Region slicer** and a **Category slicer**, both connected to all three PivotTables simultaneously via Report Connections, allowing any region or category to be isolated across every view with a single click.
- Conditional formatting on the regional summary table, providing an immediate red/green visual signal of performance against target without requiring the viewer to read exact figures.

12. Key Insights

- **East is the strongest region** on both revenue (\$44.98M) and order volume (57,034 orders), making it the benchmark region for replicable sales practices.
- **South underperforms on both revenue and order count simultaneously** (\$25.10M / 37,935 orders), pointing toward a demand or market-reach gap rather than a pricing issue — a normal order count with low revenue would have suggested pricing/discounting instead.
- **Electronics drives the largest share of revenue** (40.4%) with comparatively fewer orders than Clothing & Apparel, indicating a materially higher average order value and making it the most revenue-efficient category.
- **Accessories is significantly under-monetized** (7.1% of revenue despite a meaningful order count), representing a potential upsell/cross-sell opportunity rather than a category to scale independently.
- **California and Arizona are disproportionately strong states**, each generating over 10,000 orders — roughly double the next tier of states — suggesting state-level (not just region-level) analysis may

reveal further optimization opportunities.

13. Recommendations

- **Investigate South region's demand gap directly** — e.g., through customer/market surveys or a review of sales team coverage — before considering price-based interventions, since the data does not point to pricing as the driver.
- **Study East region's sales approach** as a potential internal best-practice model, given its lead on both revenue and order volume rather than one metric alone.
- **Explore bundling Accessories with Electronics purchases** (e.g., phone cases, charging cables alongside smartphones) to lift Accessories' revenue contribution using Electronics' stronger customer traffic.
- **Prioritize California and Arizona in account management and retention efforts**, given their outsized share of total order volume relative to all other states.
- **Standardize data entry/import processes for the Order_Date field** to prevent the mixed text/date format issue identified in Section 5 from recurring in future data exports.

14. Conclusion

This project demonstrates a complete, realistic sales analytics workflow built entirely in Microsoft Excel on a substantial 200,000-row dataset: from raw data cleaning and quality control, through conditional aggregation and lookup-based benchmarking, to PivotTable summarization and an interactive, slicer-driven dashboard. Beyond the technical execution, the analysis surfaced genuine, actionable business insights — a dual-metric underperformance in the South region, a revenue-efficiency gap between Electronics and Clothing & Apparel, and a state-level concentration effect in California and Arizona — each translated into a concrete recommendation. The project also documents a real data-quality issue (mixed date formats) that was identified and resolved methodically, illustrating the kind of attention to data integrity expected in a sales analytics or technical sales support function.

15. Tools & Skills Demonstrated

Category	Specific Skills Applied
Data Cleaning	Header correction, blank/duplicate detection, mixed-type date resolution (ISNUMBER + DATE pars
Formulas	IF, SUMIFS, COUNTIFS, VLOOKUP, XLOOKUP, AVERAGE, SUM
Data Structuring	Excel Tables (structured references), reference/lookup tables
Reporting Tools	PivotTables, PivotCharts, Value Field Settings, Value Filters (Top 10)
Visualization	Conditional Formatting (rule-based red/green flags), Bar Charts, Line Charts
Interactivity	Slicers with multi-PivotTable Report Connections
Analysis	Regional/category benchmarking, KPI definition, root-cause style interpretation of metrics

Table 8: Full skills matrix demonstrated across the project.

End of Report — B2B Sales Performance & Regional Analysis Dashboard